

Deep neural network for binary classification

With the availability of big data it makes obligatory to develop learning models that can learn every bit of important information hidden into noisy cluttered data to correctly classify the task. In order to retrieve more information from big data source we need to incorporate more parameters in the learning models and need to develop techniques to train them quickly and properly. During training process through combined interaction of these parameters we can get an optimally trained classification model that can operate with high accuracy during testing stage. Deep neural networks are one of those models that can achieve what we are aiming for through incorporation of large number of parameter sets in the model. Recent experiment in computer vision and language processing tasks showed that deep neural network can achieve state of the art result and out perform all shallow learning models. All shallow learning models we have for instance support vector machine, logistic regression, random forest or ensemble of these models can learn about near by local features and ignore long term dependency or distantly related features and they also don't have enough parameters to learn all local features if the dataset is too big. Therefore the accuracy level may go down with a larger dataset. On the contrary in deep neural network models no matter how sparsely features are distributed with in data as long as the data contains some important classification or regression information a properly trained model with appropriate optimization method will be able to learn them. It is also possible to encode the distantly related feature information. With proper optimization technique we may not even need to worry about the presence of noise in the data which is not true for other shallow learning models. For our binary classification task we decided to use convolutional neural network and recurrent neural network as both have achieved state of the art classification result for large object and text classification tasks. We want to use these models as our data has a resemblance with text data where we try to learn word vectors. And we particularly want to use these two categories of models of deep neural net because in convolutional neural net we perform convolution operation. This application of convolution encodes the sparsely distributed feature as continuous overlapped related feature information. Overlapped feature information is more useful in drawing decision leading to higher accuracy in classification result. Recurrent Neural network is well known for memorizing long term dependencies present in the big data source making it useful for designing time series problems. However it can also be used for other classification problems for instance object classification, scene understanding and binary classification. However the problem of incorporating large number of parameters into the model is learning them properly through appropriate training process. Specially we can easily overfit to our training dataset. Extracting every bit of feature information from training set does not mean it is going to be well generalized in real world data. Large parameter set also create a highly non convex object function with many local minima and stuck into any one of this local minima means the model will be unable to learn the dataset any further. We also need to be careful about the convergence time for the model. Too many parameters may make the convergence time infeasible. In order to overcome all these problems we also intend to implement all the latest development in deep neural network training mechanism. And this includes implementing drop out methodology to create combination of network parameters, implementing rectified linear units instead of sigmoid functions, RMS propagation to get dynamic learning rate that will remove the vanishing effect that severely damage the learning of parameters at the initial layers, early stopping before network start diverging from the optimum point and pretrained weight in order to get better initialization for training process. With accurate and proper implementation of latest developed techniques we can certainly get a highly accurate classification model for our side effect prediction dataset.